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Let A be an Algorithm to solve the following problem:
Given: number N
Question: Is there a number K s.t N=3k
Please describe input and output of A

The input on the algorithm A takes single integer N and outputs a Boolean True if N is a multiple of 3 else outputs false if not.

Consider the following C-Function

```
Bool Stupid(int I) {
int k = 1;
while(1) {
if(k++ == 1) break;
return I
}
}
```

Is it an Algorithm? Briefly state your reasoning. How much does K use?

Stupid it is an algorithm because it halts on all inputs, even though the return is inside the while loop. This makes it not a well defined function. Additionally, k is an single integer variable, so it uses constant memory O(1).

Mr.X describes his algorithm called foo using pseudo-code as follows
Input: Integer I
Let k be an integer variable initialized to be 1;
while(1)
if K ++ == 1 break;
Ret Yes;
Is Foo an Algorithm? Briefly State your Reasoning? How much memory does K Use?

Foo is an algorithm because it halts on all inputs. Additionally, K uses constant memory O(1)

Memory Size is measured in bits. Assume that we have 64 students in our class. Consider the following
Psuedo-Code Segment:
For Each Student in our class
If Student is Sleepy, Wake Him/Her up
What is the minimal memory size that student takes? Briefly state your reasoning

The minimal memory that student takes is 6 bits to handle 64 students in class. Because there is 64 students that means 64 states making the formula $\log_2 64 = n$ meaning $2^n = 64$, thus $n = 6$.

Let Q be an arbitrary C-Program without input. Consider the following "Code" P
Input: Q and Integer I
If Q Runs forever
return 2I
else Return I;
Is P an algorithm? Briefly state your reasoning

P is not an algorithm because of how it requires deciding whether Q halts, so P may not halt.